

MEGAN MULCAHY

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EDUCATION

Master of Business Administration, The University of Alabama

Expected May 2027

Focus Area: Strategic Management

Bachelor of Science in Computer Science, The University of Alabama

May 2026

Focus Area: Cyber Security | Minor: Mathematics | Honors College | GPA: 3.70/4.00

EXPERIENCE

Information Technology Co-op, Mercedes-Benz U.S. International, Vance, AL

Aug 2024-Dec 2025

- Engineered enterprise workflow systems integrating APIs, SQL databases, SharePoint, and Power Platform services
- Automated cross-system data pipelines, reducing manual processing workload by 60%+
- Built operational reporting systems supporting logistics, compliance, and decision workflows
- Implemented role-based access controls and audit logging across enterprise applications
- Delivered cybersecurity training to 400+ employees, translating technical concepts for business users

Lead Programming Instructor, ID Tech, Dallas, TX

May 2026-Aug 2026

- Led operations and program delivery for STEM camps serving 70+ students weekly, managing instructor teams and resolving escalated student conflict
- Instructed Python programming and AI API integration to teenage students

RELEVANT PROJECT WORK

Instagram Scraper

May 2026-Present

- Designed three-layer scraping architecture combining raw HTTP requests, Playwright browser automation, and yt-dlp for resilient media extraction
- Handled dynamic content rendering, authentication flows, and anti-bot detection using Playwright with stealth configuration
- Structured output for downstream AI pipeline ingestion, enabling multimodal content extraction and analysis

RAG Pipeline & Retrieval System

Jan 2026-June 2026

- Designed RAG pipeline using LlamaIndex with BM25 + dense vector hybrid retrieval and sentence-transformers for semantic embeddings over structured and unstructured data
- Developed FastAPI backend for document ingestion, processing workflows, and AI query services
- Implemented two-pass page classification for mixed-content PDF ingestion; OCR pipeline using Tesseract, PaddleOCR, and PyMuPDF for financial document extraction
- Evaluated retrieval quality using RAGAs framework, iterating on chunking strategy and reranking to improve precision and recall

PhD Student Data Collection Portal

Jan 2026-May 2026

- Developed full-stack data collection platform using React, Node.js, and REST APIs for structured academic workflows
- Designed PostgreSQL (Google Cloud SQL) backend for secure data storage, retrieval, and system scalability
- Containerized application using Docker for consistent deployment and environment reproducibility
- Implemented Agile development practices including sprint planning, Git workflows, and iterative feature delivery

LEADERSHIP EXPERIENCE & INVOLVEMENT

Women In Technology | Alpha Delta Pi Sorority (Homecoming Chair) | Sigma Alpha Lambda Honors Society

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, JavaScript, HTML/CSS

AI / ML: RAG, LLM APIs, Embeddings, Vector Search, LlamaIndex, LangChain, Sentence-Transformers, Whisper, OCR (Tesseract, PaddleOCR), Prompt Engineering, BM25, PyMuPDF, RAGAs

Backend: FastAPI, Node.js, REST APIs, PostgreSQL, MySQL, Docker, Linux, Google Cloud Platform

Tools & Foundations: Playwright, yt-dlp, ChromaDB, React, Power Platform, System Design, Compilers & Interpreters